Business Finance - Written Exam Q&A

117 model answers by unit, with marks, marking scheme and where each question was asked

How to use: 2 marks = definition + 1 point (3-4 lines) · 3 marks = intro + 3 points · 5-6 marks = intro + 5-6 headed points + example/table · 10 marks = intro, headings, table/diagram, example, conclusion. **Stars:** *** asked in your PDFs (very likely), ** core syllabus, * possible. Answers marked [answer from standard knowledge] are not in your notes - check them with class notes.

Unit I

Q1. [2 marks] Define the finance function of a business.

Finance function · ***

The **finance function** is that part of management which is concerned with the **procurement** of funds and their **effective utilisation** for achieving the firm's objectives.

- It covers four decisions: investment, financing, dividend and liquidity.
- Every business decision has financial implications.

Marking scheme: Definition 1 + point 1

Asked in: Assignment 1 (TVM) Q1

Q2. [3 marks] Explain the finance function and state its two broad decision areas.

Finance function · ***

The finance function is procurement and effective utilisation of funds. Its broad decision areas:

1. **Investment decision** - where and how much to invest (capital budgeting, working capital).
2. **Financing decision** - how much finance to raise and from which sources (debt-equity mix).

Dividend and liquidity decisions support these two.

Marking scheme: Meaning 1 + two areas 2

Asked in: Assignment 1 (TVM) Q1

Q3. [3 marks] List the other functions of finance (apart from the major decisions).

Finance function · **

1. Estimating the requirement of funds (forecasting / planning).
2. Determining the sources of funds - long and short term.
3. Allocation and efficient utilisation of funds.
4. Financial control - ratios, budgets and audits.
5. Profit planning and management of surplus.

Marking scheme: 5 × ⅗

Q4. [6 marks] Explain the finance function and its decision areas with examples.

Finance function · **

Decision	Meaning	Example
Investment	Where and how much to invest in long-term and short-term assets	Buying new machinery
Financing	How much to raise and from which sources	Debt-equity ratio; shares vs bank loan
Dividend	How much profit to pay out vs retain	Declaring 20% dividend, retaining the rest

Liquidity (working capital)	Manage current assets without shortage or idle funds	Keeping cash to pay suppliers

- Other functions: estimating funds requirement, determining sources, allocation of funds, financial control, profit planning.

Marking scheme: Meaning 1 + 4 decisions with examples 4 + other functions 1

Q5. [2 marks] What are owners' funds? Give two sources.

Funds · **

Capital contributed by the owners (proprietors, partners, shareholders) plus profits retained in the business. Owners bear the **residual risk**.

- Sources: equity shares, preference shares, retained earnings / reserves.

Marking scheme: Meaning 1 + sources 1

Q6. [3 marks] Distinguish between owners' funds and borrowed funds (any three points).

Funds · ***

Basis	Owners' funds	Borrowed funds
Source	Equity, preference shares, retained earnings	Debentures, bonds, term loans
Return	Dividend, only out of profits	Fixed interest, whatever the profit
Repayment	Not repayable during life	Repayable after a fixed period
Risk	Higher	Lower
Cost	Dividend from post-tax profit	Interest is tax-deductible

Marking scheme: 3 points × 1

Q7. [6 marks] Distinguish between owners' funds and borrowed funds.

Funds · ***

Basis	Owners' funds	Borrowed funds
Meaning	Provided by owners from own resources	Arranged from outsiders - banks, FIs, debenture holders
Sources	Equity, preference, retained earnings	Debentures, bonds, term loans, public deposits, trade credit
Ownership	Provider is part-owner	Provider is only a creditor
Control	Voting rights and control	No control except on default
Return	Dividend, only from profit, not guaranteed	Fixed interest irrespective of profit
Repayment	Not repayable in lifetime	Repayable after fixed period
Security	None needed	Usually a charge on assets
Risk	Higher	Lower
Cost	Higher (dividend from post-tax profit)	Lower (interest is tax-deductible)
Effect on structure	Strengthens equity base	Raises leverage / gearing

Marking scheme: 6 points × 1

Q8. [10 marks] Explain owners' funds and borrowed funds. Discuss their sources, merits, demerits and how a firm decides the right mix.

Funds · ***

Owners' funds

- Capital of owners + retained profits (reserves and surplus).
- Sources: equity shares, preference shares, retained earnings.
- Owners bear residual risk; no fixed obligation to pay dividend or repay capital.

Borrowed funds

- Raised from banks, institutions and the public.
- Sources: debentures, bonds, term loans, public deposits, trade credit.
- Fixed interest, repayable, usually secured.

Comparison

Basis	Owners' funds	Borrowed funds
Ownership & control	Part-owner, voting rights	Creditor, no control
Return	Dividend, not guaranteed	Interest, fixed
Repayment	Not repayable	Repayable
Risk	Higher	Lower
Cost	Higher	Lower (tax-deductible)
Effect	Strengthens equity	Raises leverage

Choosing the mix

- Compare cost, risk, control and flexibility.
- Check cash-flow ability to service interest and principal.
- Use EBIT-EPS analysis; above the indifference EBIT debt is better.
- Pecking order: internal funds → debt → hybrid → new equity.

Example: A firm earning more than its cost of debt gains from trading on equity.

Marking scheme: Owners 2 + borrowed 2 + comparison 3 + mix 3

Q9. [10 marks] Explain the meaning, functions and objectives of financial management. Why is wealth maximisation preferred?

Introduction · **

Meaning

Business finance = acquiring funds and using them efficiently to achieve the objectives of the enterprise.

Functions

- Investment, financing, dividend and liquidity decisions.
- Estimating funds requirement; determining sources; allocation of funds.
- Financial control and profit planning.

Objectives

- **Profit maximisation** - criticised: ambiguous, ignores time and risk.
- **Wealth maximisation** - maximise market value of shares; considers time, risk, magnitude and timing of benefits.

Why wealth maximisation?

- Precise and measurable.
- Considers time value of money and risk.
- Reflects the true economic value of the firm.

Owners' vs borrowed funds

Owners' funds carry no fixed obligation but bear more risk; borrowed funds are cheaper (tax-deductible interest) but increase leverage.

Marking scheme: Meaning 1.5 + functions 3 + objectives 2.5 + reasons 3

Q10. [2 marks] What is wealth maximisation?

Objectives · **

Wealth maximisation (value maximisation) means maximising the **market value of the firm's equity shares**, i.e.

shareholders' wealth.

```
Shareholders' wealth = Number of shares held × Market price per share
```

It considers time value of money and risk, so it is the accepted goal of financial management.

Marking scheme: Meaning 1 + formula/reason 1

Q11. [3 marks] Distinguish between profit maximisation and wealth maximisation (any three points).

Objectives · ***

Basis	Profit maximisation	Wealth maximisation
Focus	Short-term earnings	Long-term market value
Time value	Ignored	Considered by discounting
Risk	Ignored	Considered
Clarity	Vague (EPS? total profit?)	Precise

Conclusion: wealth maximisation is the correct goal.

Marking scheme: 3 points × 1

Asked in: Assignment 1 (TVM) Q2

Q12. [3 marks] State the arguments for and against profit maximisation.

Objectives · ***

In favour	Criticism
Efficient allocation of resources	Ambiguous - total profit, rate of profit or EPS?
Easy to measure and understand	Ignores timing of returns (time value of money)
Ensures welfare of shareholders and society	Ignores risk and quality of benefits

Marking scheme: 3 + 3 points (½ each)

Q13. [6 marks] Distinguish between profit maximisation and wealth maximisation as objectives of financial management.

Objectives · ***

Profit maximisation is the traditional goal; wealth maximisation (maximising the market value of shares) is the modern goal.

Basis	Profit maximisation	Wealth maximisation
Meaning	Earning maximum profit	Maximising market value of equity shares
Focus	Short-term earnings	Long-term value
Time value of money	Ignored	Considered (discounting)
Risk	Ignored	Considered
Quality of benefits	Not distinguished	Certainty considered
Measurement	Absolute profit / EPS	Market price of share / NPV
Clarity	Ambiguous	Clear and precise

Conclusion: wealth maximisation is precise, considers time and risk, and is the universally accepted goal.

Marking scheme: 6 points × 1 (conclusion adds to any point)

Asked in: Assignment 1 (TVM) Q2

Q14. [10 marks] Explain the objectives of financial management. Distinguish between profit maximisation and wealth maximisation and state why wealth maximisation is the better goal.

Objectives · ***

Objectives

- **Profit maximisation** - the traditional approach; profit is a test of efficiency.
- **Wealth maximisation** - maximise the market value of shares: wealth = shares held × market price.
- **Supporting:** optimum use of funds, adequate and regular supply, safety of funds, sound capital structure, survival and growth.

Profit maximisation - arguments and criticisms

- Efficient resource allocation; simple to measure.
- Ambiguous; ignores time value; ignores risk; ignores quality of benefits.

Comparison

Basis	Profit maximisation	Wealth maximisation
Focus	Short-term earnings	Long-term market value
Time value	Ignored	Considered
Risk	Ignored	Considered
Measurement	Profit / EPS	Market price / NPV
Clarity	Vague	Precise

Why wealth maximisation?

It is precise, includes time value and risk, and shows the true economic value of the firm. **Conclusion:** it is the accepted goal of financial management.

Marking scheme: Objectives 3 + criticisms 2 + comparison 3 + conclusion 2

Asked in: Assignment 1 (TVM) Q2

Unit II

Q15. [2 marks] What is an annuity? Give two examples.

Annuity · **

An **annuity** is a series of **equal cash flows at regular intervals** for a specified number of periods.

- Examples: EMIs on a loan, insurance premiums, equal yearly deposits.

Marking scheme: Definition 1 + examples 1

Q16. [2 marks] What is a perpetuity? Write its formula.

Annuity · **

A **perpetuity** is an annuity that continues for ever.

```
PV of perpetuity = A ÷ i
```

Example: ₹500 a year for ever at 10% → PV = 500 ÷ 0.10 = ₹5,000.

Marking scheme: Meaning 1 + formula 1

Q17. [2 marks] What is an annuity due?

Annuity · **

An annuity whose payments are made at the **beginning** of each period.

```
Value (due) = Value (ordinary) × (1 + i)
```

Example: Ordinary FVA ₹22,530 at 8% → due = 22,530 × 1.08 = ₹24,332.

Marking scheme: Meaning 1 + formula 1

Q18. [10 marks] Explain annuities: future value, present value, perpetuity and annuity due, with formulae and numerical examples.

Annuity · ***

Meaning

A series of equal cash flows at regular intervals (EMIs, premiums, equal deposits).

Type	Formula	Example
FV of annuity	$A[((1+i)^n-1)/i]$	₹5,000 × 4.5061 (4 yrs, 8%) = ₹22,530
PV of annuity	$A[(1-(1+i)^{-n})/i]$	₹1,000 × 2.4869 (3 yrs, 10%) = ₹2,487
Perpetuity	A ÷ i	₹500 ÷ 0.10 = ₹5,000
Annuity due	Ordinary × (1 + i)	22,530 × 1.08 = ₹24,332

Uses

- FVA - savings plans, recurring deposits, sinking funds.
- PVA - NPV with equal inflows, loan eligibility.
- Perpetuity - irredeemable bonds, preference shares.
- Due - rentals and advance premiums.

Loan amortisation (application of PVA)

```
A = Loan ÷ PVIFA(i, n); ₹1,00,000 at 10% for 5 years → ₹26,380
```

Interest = opening balance × i; principal = instalment – interest.

Marking scheme: Meaning 1 + table 4 + uses 2 + amortisation 3

Q19. [6 marks] Explain the future value and present value of an annuity with examples.

Annuity & loan · **

Future value of an annuity

$$FVA = A \times [((1+i)^n - 1) \div i]$$

Example: ₹5,000 at the end of each year for 4 years at 8%: FVA = 5,000 × 4.5061 = **₹22,530**.

Present value of an annuity

$$PVA = A \times [(1 - (1+i)^{-n}) \div i]$$

Example: ₹1,000 a year for 3 years at 10%: PVA = 1,000 × 2.4869 = **₹2,487**.

Annuity due: value = ordinary value × (1+i).

Marking scheme: Each formula 1.5 + example 1.5

Q20. [3 marks] ₹5,000 is deposited at the beginning of each year for 4 years at 8%. Find the future value.

Annuity due · **

Ordinary annuity: FVA = 5,000 × 4.5061 = ₹22,530

```
Annuity due = 22,530 × (1 + 0.08)
```

≈ ₹24,332

Marking scheme: Ordinary 1.5 + due factor 1.5

Q21. [3 marks] Mr Arjun deposits ₹50,000 at 8% p.a. compounded annually. Find the terminal value after 4 years.

Compound value · ***

$$FV = PV \times (1 + i)^n = 50{,}000 \times (1.08)^4$$

$(1.08)^4 = 1.3605$

FV = 50,000 × 1.3605 = ₹68,024

Marking scheme: Formula 1 + factor 1 + answer 1

Asked in: Assignment 1 (TVM) Q4

Q22. [3 marks] ₹10,000 is invested at 12% p.a. compounded quarterly for 2 years. Find the future value.

Compounding · **

m = 4 → rate per quarter = 12% ÷ 4 = 3%; periods = 2 × 4 = 8

$$FV = 10{,}000 \times (1.03)^8 = 10{,}000 \times 1.2668$$

FV ≈ ₹12,668

Marking scheme: Rate/periods 1 + formula 1 + answer 1

Q23. [6 marks] Calculate: (a) the future value of ₹20,000 at 12% for 3 years; (b) the present value of ₹30,000 receivable after 4 years at 10%; (c) the present value of ₹2,000 a year for 5 years at 8% (PVIFA = 3.9927).

FV, PV and annuity · ***

(a) $FV = 20{,}000 \times (1.12)^3 = 20{,}000 \times 1.4049 =$ **₹28,099**

(b) $PV = 30{,}000 \div (1.10)^4 = 30{,}000 \div 1.4641 =$ **₹20,490**

(c) $PVA = 2{,}000 \times 3.9927 =$ **₹7,985**

Each part: write the formula, substitute, give the answer with ₹.

Marking scheme: 3 × 2 (formula 0.5 + working 1 + answer 0.5)

Q24. [2 marks] What is loan amortisation?

Loan · **

Repaying a loan (principal + interest) through a series of **equal periodic instalments** calculated with the present-value-of-annuity formula, so the balance falls to zero at the end of the term.

$$A = \text{Loan} \div \text{PVIFA}(i, n)$$

Marking scheme: Meaning 1.5 + formula 0.5

Q25. [3 marks] A loan of ₹1,00,000 at 10% is repaid in 3 equal annual instalments. Find the instalment and the interest of year 1.

Loan · **

$$A = \text{Loan} \div \text{PVIFA}(10\%, 3) = 1{,}00{,}000 \div 2.4869$$

A ≈ ₹40,212

Year 1 interest = 1,00,000 × 10% = ₹**10,000** (principal repaid = 40,212 – 10,000 = ₹30,212).

Marking scheme: Formula 1 + instalment 1 + interest 1

Q26. [6 marks] A loan of ₹1,00,000 at 10% is repaid in 3 equal annual instalments. Prepare the amortisation schedule.

Loan · **

```
A = 1,00,000 ÷ 2.4869 ≈ ₹40,212
```

Year	Opening	Instalment	Interest (10%)	Principal	Closing
1	1,00,000	40,212	10,000	30,212	69,788
2	69,788	40,212	6,979	33,233	36,555
3	36,555	40,212	3,656	36,556	≈ 0

Interest each year = opening balance × 10%; principal = instalment – interest.

Marking scheme: Instalment 1.5 + 3 rows × 1.5

Q27. [3 marks] Using the PVF table, find the present value of ₹50,000 receivable after 3 years at 10%. (PVF year 3 = 0.751)

Present value · ***

```
PV = FV × PVF = 50,000 × 0.751
```

PV = ₹37,550

Check: 50,000 ÷ $(1.10)^3$ = 37,566 (table factor rounds to 0.751).

Marking scheme: Time value explained 1 + working 1 + answer 1

Asked in: Online Preparatory Test (30-09-2026) Q2

Q28. [3 marks] Explain the relationship between present value and the discount rate with an example.

Present value · ***

They are **inversely related**: when the discount rate rises, the present value of a future sum falls.

Example: PV of ₹13,310 after 3 years: at 10% → ₹10,000; at 15% → 13,310 ÷ 1.5209 = ₹8,751. Higher rate → lower PV.

- Rate = 0 → PV equals the future value.
- Longer period → PV falls further.

Marking scheme: Relationship 1 + example 2

Asked in: Assignment 1 (TVM) Q3

Q29. [2 marks] What is the time value of money?

Time value of money · ***

Time value of money means that a rupee received **today is worth more** than a rupee received in the future, because today's money can be invested to earn a return.

- Reasons: opportunity to earn interest, risk and uncertainty, inflation, preference for liquidity.

Marking scheme: Meaning 1 + reasons 1

Asked in: Online Preparatory Test (30-09-2026) Q2

Q30. [2 marks] Distinguish between compounding and discounting.

Time value of money · **

Basis	Compounding	Discounting

Direction	Present → future	Future → present
Result	Future (terminal) value	Present value
Formula	$FV = PV(1+i)^n$	$PV = FV/(1+i)^n$

Marking scheme: 2 differences × 1

Q31. [3 marks] Why does money have a time value? Give reasons.

Time value of money · ***

1. **Opportunity cost** - money available now can be invested to earn interest.
2. **Risk and uncertainty** attached to future receipts.
3. **Inflation** reduces purchasing power over time.
4. **Liquidity preference** - people prefer current consumption to future consumption.

Marking scheme: 3-4 reasons × 1

Q32. [6 marks] Explain the time value of money and the relationship between present value and the discount rate with an example.

Time value of money · ***

Concept: a rupee today is worth more than a rupee in future - opportunity cost, risk, inflation, liquidity preference.

- **Compounding:** $FV = PV(1+i)^n$
- **Discounting:** $PV = FV/(1+i)^n$

Discount rate	PV of ₹13,310 after 3 years
0%	₹13,310 (equals future value)
10%	₹10,000
15%	₹8,751

Higher rate → lower PV (inverse, non-linear). Longer period → even lower PV.

Marking scheme: Concept 1.5 + formulas 1.5 + table 2 + conclusion 1

Asked in: Assignment 1 (TVM) Q3

Q33. [10 marks] Explain the time value of money. Discuss compounding, present value, annuities and loan amortisation with examples.

Time value of money · ***

1. Concept

A rupee today is worth more than a rupee tomorrow (opportunity cost, risk, inflation, liquidity preference).

2. Compounding

$FV = PV(1+i)^n$; with m compoundings: $PV(1+i/m)^{nm}$

Example: ₹10,000 at 10% for 3 years = ₹13,310.

3. Present value

$PV = FV \div (1+i)^n$

Inverse relation with the discount rate.

4. Annuities

$FVA = A[((1+i)^n-1)/i]$ $PVA = A[(1-(1+i)^{-n})/i]$ $Perpetuity = A/i$

5. Loan amortisation

```
A = Loan ÷ PVIFA(i,n)
```

Year	Opening	Instalment	Interest	Closing
1	1,00,000	40,212	10,000	69,788
2	69,788	40,212	6,979	36,555
3	36,555	40,212	3,656	0

Marking scheme: Concept 1.5 + compounding 2 + PV 2 + annuity 2.5 + amortisation 2

Asked in: Assignment 1 (TVM)

Unit III

Q34. [3 marks] A bond of face value ₹1,000 has coupon 9% (annual), 3 years to maturity; required return 10%. Find its value.

Bond value · ***

```
V = I × PVIFA(10%,3) + F × PVIF(10%,3)
```

I = 9% × 1,000 = ₹90

V = 90 × 2.4869 + 1,000 × 0.7513 = 223.82 + 751.31

V ≈ ₹975 (a discount, because coupon 9% < yield 10%).

Marking scheme: Formula 1 + working 1.5 + answer/comment 0.5

Asked in: Assignment 1 (TVM) Q5(a)

Q35. [3 marks] A zero-coupon bond has face value ₹1,000 and 2 years to maturity. Required return is 10%. Find its value.

Bond value · **

```
V = F ÷ (1 + kd)^n = 1,000 ÷ (1.10)^2 = 1,000 ÷ 1.21
```

V ≈ ₹826.45

Marking scheme: Formula 1 + working 1 + answer 1

Q36. [3 marks] A perpetual bond pays ₹100 interest every year; required return is 8%. Find its value.

Bond value · **

```
V = I ÷ kd = 100 ÷ 0.08
```

V = ₹1,250

Marking scheme: Formula 1.5 + answer 1.5

Q37. [3 marks] A ₹1,000 bond pays 10% interest half-yearly for 2 years; yield is 10%. Find its value.

Bond value · **

Half-yearly interest = ₹50; rate = 5%; periods = 4

```
V = 50 × PVIFA(5%,4) + 1,000 × PVIF(5%,4) = 50 × 3.5460 + 1,000 × 0.8227
```

= 177.3 + 822.7 = **₹1,000** (coupon = yield → par).

Marking scheme: Adjusting rate/periods 1 + working 1.5 + comment 0.5

Q38. [6 marks] A bond of face value ₹1,000 pays 8% interest annually, has 4 years to maturity, and investors require 10%. Find its value and state whether it sells at a premium or discount. (PVIFA 10%,4 = 3.1699; PVIF 10%,4 = 0.6830)

Bond value · ***

I = 8% × 1,000 = ₹80

```
V = 80 × 3.1699 + 1,000 × 0.6830 = 253.59 + 683.00
```

V ≈ ₹936.6

Coupon (8%) < yield (10%) → the bond sells at a **discount** (value below ₹1,000). If yield rose further, value would fall (inverse relationship).

Marking scheme: Interest 1 + formula 1.5 + working 1.5 + answer 1 + comment 1

Q39. [2 marks] State the relationship between coupon rate, market yield and bond price.

Bonds · **

Condition	Bond sells at
Coupon = Yield	Par
Coupon > Yield	Premium
Coupon < Yield	Discount

Bond value and yield move in opposite directions.

Marking scheme: Table 1.5 + inverse relation 0.5

Q40. [2 marks] What are the assumptions of the Gordon growth model?

Gordon model · ***

1. Dividends grow at a **constant rate g for ever**.
2. Required return **ke > g** (otherwise the value is negative / meaningless).
3. Value is the present value of all future dividends.

Marking scheme: Each assumption 1 (any 2)

Q41. [3 marks] A share is priced at ₹80; dividend just paid ₹4 grows at 6% for ever. Find the investor's expected return.

Gordon model · ***

```
ke = D1 ÷ P0 + g
```

D_1 = 4 × 1.06 = ₹4.24 → dividend yield = 4.24 ÷ 80 = 5.3%

ke = 5.3% + 6% = 11.3%

Marking scheme: D1 1 + formula 1 + answer 1

Asked in: Assignment 1 (TVM) Q5(b)

Q42. [3 marks] D_0 = ₹4, growth 5%, required return 15%. Find the share price.

Gordon model · ***

```
P0 = D1 ÷ (ke − g)
```

D_1 = 4 × 1.05 = 4.20

P_0 = 4.20 ÷ (0.15 − 0.05) = ₹42

Marking scheme: D1 1 + formula 1 + answer 1

Q43. [6 marks] D_0 = ₹5, g = 4%, ke = 12%. (a) Find the value of the share. (b) If the market price is ₹50, find the expected return.

Gordon model · ***

$D_1 = 5 \times 1.04 = ₹5.20$

```
(a) P0 = 5.20 ÷ (0.12 − 0.04) = ₹65

(b) ke = D1/P0 + g = 5.20 ÷ 50 + 0.04 = 10.4% + 4% = 14.4%
```

At ₹50 the share is undervalued (value ₹65, price ₹50) - the expected return 14.4% is above the required 12%.

Marking scheme: D1 1 + (a) 2 + (b) 2 + comment 1

Q44. [3 marks] D_1 = ₹5, expected price after one year ₹110, required return 15%. Find the value of the share today.

Share valuation · **

```
P0 = D1/(1+ke) + P1/(1+ke) = (5 + 110) ÷ 1.15
```

P_0 = ₹100

Marking scheme: Formula 1 + working 1 + answer 1

Q45. [3 marks] A share bought for ₹100 gave a dividend of ₹8 and was sold for ₹112. Find the rate of return.

Share valuation · ***

```
Return = [D + (P1 − P0)] ÷ P0 = [8 + (112 − 100)] ÷ 100
```

= 20% (dividend yield 8% + capital gain 12%)

Marking scheme: Formula 1 + working 1 + answer 1

Q46. [6 marks] Explain the methods of valuing a bond with the relationship between value, coupon rate and yield.

Valuation · **

- **Redeemable bond (annual interest):** V = I × PVIFA(kd,n) + F × PVIF(kd,n)
- **Semi-annual interest:** V = (I/2) × PVIFA(kd/2, 2n) + F × PVIF(kd/2, 2n)
- **Perpetual bond:** V = I ÷ kd
- **Zero-coupon bond:** V = F ÷ $(1+kd)^n$

Example: Face 1,000, coupon 10%, 3 years, required return 12% → V = 100 × 2.402 + 1,000 × 0.7118 = ₹952 (discount).

Coupon vs yield	Price
Coupon = yield	Par
Coupon > yield	Premium
Coupon < yield	Discount

Marking scheme: Four formulas 2.5 + example 1.5 + relationship 2

Q47. [10 marks] Explain the valuation of bonds and stocks. Illustrate with the Gordon growth model.

Valuation · ***

Bonds

- $V = I \times PVIFA(kd,n) + F \times PVIF(kd,n)$
- Perpetual: $V = I/kd$; Zero coupon: $V = F/(1+kd)^n$

Example: Face 1,000, coupon 9%, 3 yrs, required return 10% → ₹975 (discount).

Stocks

- Single period: $P_0 = D_1/(1+ke) + P_1/(1+ke)$
- No growth: $P_0 = D/ke$
- Return: $[D_1 + (P_1 - P_0)]/P_0$

Gordon growth model

$$P_0 = D_1 \div (ke - g),\ D_1 = D_0(1+g) \cdot ke = D_1/P_0 + g$$

Example: D_0 = ₹4, g = 5%, ke = 15% → P_0 = ₹42. Valid only if ke > g.

Marking scheme: Bond formulas 3 + example 1.5 + stock formulas 2 + Gordon 3.5

Unit IV

Q48. [2 marks] What is the Accounting Rate of Return (ARR)? State its rule.

ARR · ***

```
ARR = (Average annual PAT ÷ Average investment) × 100
```

Accept the project if ARR is **greater than the cut-off rate**; among alternatives choose the highest ARR.

Marking scheme: Formula 1 + rule 1

Asked in: Online Preparatory Test (30-09-2026) Q4

Q49. [3 marks] Machine costs ₹1,20,000, scrap ₹20,000, life 4 years; PAT: 20,000, 25,000, 30,000, 25,000. Calculate ARR on average investment.

ARR · ***

Average PAT = (20 + 25 + 30 + 25) ÷ 4 = ₹25,000

Average investment = (1,20,000 – 20,000) ÷ 2 + 20,000 = ₹70,000

```
ARR = 25,000 ÷ 70,000 × 100
```

ARR = 35.71%

Marking scheme: Average PAT 1 + average investment 1 + ARR 1

Asked in: Online Preparatory Test (30-09-2026) Q4

Q50. [3 marks] Machine ₹4,00,000, scrap ₹40,000, average PAT ₹66,000. Calculate ARR on average investment.

ARR · ***

Average investment = (4,00,000 – 40,000) ÷ 2 + 40,000 = ₹2,20,000

```
ARR = 66,000 ÷ 2,20,000 × 100
```

ARR = 30%

Marking scheme: Avg investment 1.5 + ARR 1.5

Asked in: Assignment 2 (Capital Budgeting) Q6

Q51. [2 marks] Define capital budgeting.

Capital budgeting · ***

Capital budgeting is the process of evaluating and selecting **long-term investment proposals** consistent with the objective of maximising shareholders' wealth. Funds are committed now for returns spread over several years.

Marking scheme: Definition 2

Asked in: Assignment 2 (Capital Budgeting) Q1; Online Preparatory Test (30-09-2026) Q1

Q52. [2 marks] What is the payback period? State its decision rule.

Capital budgeting · **

Payback period is the **time required to recover the initial investment** from the net cash inflows.

- Rule: accept the project if payback is **less than the standard period**; among alternatives choose the shorter payback.

Marking scheme: Meaning 1 + rule 1

Q53. [2 marks] What is IRR? State its decision rule.

Capital budgeting · **

Internal Rate of Return is the discount rate at which the present value of cash inflows equals the present value of outflows, i.e. **NPV = 0**.

- Rule: accept if IRR > cost of capital; reject if IRR < cost of capital.

Marking scheme: Meaning 1 + rule 1

Q54. [2 marks] What is the profitability index? State the decision rule.

Capital budgeting · **

```
PI = PV of cash inflows ÷ Initial investment
```

Accept the project if **PI > 1** (equivalent to NPV > 0).

[extra: used in the test paper, not defined in the notes]

Marking scheme: Formula 1 + rule 1

Asked in: Online Preparatory Test (30-09-2026) Q6(b)

Q55. [3 marks] State any three features of capital budgeting decisions.

Capital budgeting · **

1. **Long-term commitment** of funds, irreversible or reversible only at a heavy loss.
2. **Substantial funds** involved; affects future profitability and competitiveness.
3. **Uncertain benefits** spread over many years, so there is risk.
4. **Strategic impact** on the growth of the firm.

Marking scheme: 3 features × 1

Q56. [6 marks] Explain the nature, importance and process of capital budgeting.

Capital budgeting · ***

Nature

- Long-term and substantial commitment of funds.
- Irreversible or reversible only at a heavy loss.
- Future benefits are uncertain - risk.
- Strategic effect on growth.

Importance

- Large investment - errors are costly.

- Long-term effect on profitability and risk.
- Irreversibility.
- Complex forecasting of distant cash flows.

Process

1. Identify proposals
2. Estimate cash flows (initial outlay, annual flows, terminal flow)
3. Evaluate - payback, ARR, NPV, IRR
4. Select the project
5. Implement and review (post-audit)

Marking scheme: Nature 2 + importance 2 + process 2

Asked in: Online Preparatory Test Q1; Assignment 2 Q1

Q57. [10 marks] What is capital budgeting? Explain its nature, importance and the techniques of appraisal (payback, ARR, NPV, IRR) with decision rules, merits and demerits.

Capital budgeting · ***

Introduction

Capital budgeting is evaluating and selecting long-term investment proposals consistent with maximising shareholders' wealth.

Nature

- Long-term, large, mostly irreversible commitment.
- Future benefits uncertain → risk.
- Strategic impact on growth.

Process

1. Identify proposals
2. Estimate cash flows
3. Evaluate (Payback, ARR, NPV, IRR)
4. Select project
5. Implement and review (post-audit)

Techniques

Method	Time value?	Rule	Merit	Demerit
Payback	No	Accept if < standard	Simple, liquidity	Ignores later flows
ARR	No	Accept if > cut-off	Uses profit data, whole life	Profit not cash
NPV	Yes	Accept if NPV > 0	Wealth maximisation	Needs accurate k
IRR	Yes	Accept if IRR > k	Percentage return	Tedious, multiple IRRs

Example: Project X: NPV = +₹40,610 at 10% and PI 1.20 → accept.

Conclusion

Discounted cash flow methods are superior; when NPV and IRR conflict, prefer NPV.

Marking scheme: Intro 1 + nature & importance 2 + process 1 + table 4 + example 1 + conclusion 1

Asked in: Online Preparatory Test (30-09-2026) Q1, Q5; Assignment 2 (Capital Budgeting)

Q58. [10 marks] Outlay ₹10,00,000; inflows ₹3,00,000, ₹3,50,000, ₹4,00,000, ₹2,50,000; cost of capital 12%. Compute payback period, NPV and IRR (by interpolation using 10% and 12%) and advise.

Capital budgeting numerical · ***

Year	Inflow	Cumulative	PVF 12%	PV 12%
1	3,00,000	3,00,000	0.8929	2,67,870

2	3,50,000	6,50,000	0.7972	2,79,020
3	4,00,000	10,50,000	0.7118	2,84,720
4	2,50,000	13,00,000	0.6355	1,58,875

```
Payback = 2 + 3,50,000 ÷ 4,00,000 = 2.875 years

NPV = 9,90,485 − 10,00,000 = −₹9,515

NPV at 10% = +33,240; IRR = 10 + [33,240 ÷ (33,240+9,515)] × 2 = ≈ 11.55%
```

Advice: payback is acceptable if the standard is longer than 2.9 years, but NPV < 0 and IRR (11.55%) < 12% → **reject**. Discounted methods decide.

Marking scheme: Payback 2 + PV table 3 + NPV 1.5 + IRR 2.5 + advice 1

Asked in: Assignment 2 (Capital Budgeting) Q4-Q5

Q59. [3 marks] State the merits and demerits of the IRR method.

IRR · ***

Merits	Demerits
Considers time value and the whole life of the project	Tedious trial and error without a calculator or spreadsheet
Expressed as a % - easy to compare with the cost of capital	Multiple or no IRR when cash-flow signs change more than once
	Assumes reinvestment at the IRR itself (unrealistic)

Marking scheme: 3 + 3 points (½ each)

Asked in: Online Preparatory Test Q5

Q60. [3 marks] Why is capital budgeting important? Explain any two reasons.

Importance of capital budgeting · ***

1. **Large investment of funds** - big amounts are committed, so errors are costly and hard to reverse.
2. **Long-term effect** - decisions affect profitability, risk and competitive position for many years.

Also: irreversibility and complexity of forecasting distant cash flows.

Marking scheme: Definition 1 + 2 reasons × 1

Asked in: Online Preparatory Test (30-09-2026) Q1; Assignment 2 (Capital Budgeting) Q1

Q61. [3 marks] A project costs ₹1,00,000 and gives ₹40,000 a year for 4 years. Cost of capital is 10%. Find NPV and say whether to accept. (PVIFA 10%, 4 yrs = 3.1699)

NPV · ***

```
PV of inflows = 40,000 × 3.1699 = ₹1,26,796

NPV = 1,26,796 − 1,00,000 = +₹26,796
```

NPV is positive → **accept**. (PI = 1.27)

Marking scheme: PV 1.5 + NPV 1 + decision 0.5

Q62. [3 marks] State the merits and demerits of the NPV method.

NPV · ***

Merits	Demerits
Considers time value of money	Needs an accurate estimate of the cost of capital
Considers the entire stream of cash flows	Absolute NPV is hard to compare for projects of different size or life
Consistent with wealth maximisation (NPV = addition to shareholders' wealth)	

Marking scheme: 3 + 3 points (½ each)

Asked in: Online Preparatory Test Q5

Q63. [6 marks] Project X costs ₹2,00,000, life 4 years, cost of capital 10%. Inflows: 80,000; 90,000; 70,000; 60,000. (a) Calculate NPV. (b) Calculate the profitability index and advise on acceptance. (PVF: 0.909, 0.826, 0.751, 0.683)

NPV · ***

Year	Inflow	PVF @10%	PV
1	80,000	0.909	72,720
2	90,000	0.826	74,340
3	70,000	0.751	52,570
4	60,000	0.683	40,980
Total PV			**2,40,610**

(a) NPV = 2,40,610 − 2,00,000 = **+₹40,610**

(b) PI = PV of inflows ÷ Investment = 2,40,610 ÷ 2,00,000 = **1.20**

Advice: NPV > 0 and PI > 1 → **accept** the project.

Marking scheme: PV table 2 + NPV 2 + PI 1 + advice 1

Asked in: Online Preparatory Test (30-09-2026) Q6

Q64. [6 marks] A project needs ₹1,00,000 and gives ₹40,000 a year for 4 years; cost of capital 10%; standard payback 3 years. Calculate payback, NPV and PI, and advise.

NPV · ***

Payback = 1,00,000 ÷ 40,000 = **2.5 years** (< 3 → acceptable)

PV of inflows = 40,000 × 3.1699 = ₹1,26,796; NPV = **+₹26,796**

PI = 1,26,796 ÷ 1,00,000 = **1.27**

Advice: payback, NPV and PI all support the project → **accept**.

Marking scheme: Payback 1.5 + NPV 2 + PI 1.5 + advice 1

Q65. [6 marks] Two projects each cost ₹1,00,000. A gives ₹50,000 a year for 3 years. B gives ₹20,000, ₹40,000 and ₹80,000 in years 1-3. Cost of capital 10% (PVIFA 3 yrs = 2.4869; PVF: 0.9091, 0.8264, 0.7513). Calculate NPV and payback and select one.

NPV · ***

Project A: PV = 50,000 × 2.4869 = 1,24,345 → **NPV = +24,345**; payback = 1,00,000 ÷ 50,000 = **2 years**

Project B: PV = 20,000×0.9091 + 40,000×0.8264 + 80,000×0.7513 = 18,182 + 33,056 + 60,104 = 1,11,342 → **NPV = +11,342**; payback = 2 + 40,000/80,000 = **2.5 years**

Decision: both are acceptable (NPV > 0) but they are mutually exclusive, so select **Project A** - higher NPV and faster payback.

Marking scheme: A 2.5 + B 2.5 + decision 1

Q66. [6 marks] Explain the NPV and IRR methods with their decision rules. State one advantage and one limitation of each.

NPV & IRR · ***

1. Net Present Value (NPV)

$NPV = \Sigma\ CF_t \div (1+k)^t - \text{Initial investment}$

- **Rule:** accept if NPV > 0; reject if NPV < 0; among exclusive projects choose the highest positive NPV.
- **Advantage:** considers time value of money and the whole cash-flow stream; matches wealth maximisation.
- **Limitation:** needs an accurate cost of capital; absolute NPV is hard to compare for projects of different size or life.

2. Internal Rate of Return (IRR)

$$IRR = LR + [NPV_{LR} \div (NPV_{LR} - NPV_{HR})] \times (HR - LR)$$

- **Rule:** accept if IRR > cost of capital (k).
- **Advantage:** expressed as a %, easy to compare with the cost of capital.
- **Limitation:** tedious trial and error; may give multiple IRRs with unconventional cash flows; assumes reinvestment at IRR.

Conflict: for mutually exclusive projects prefer NPV.

Marking scheme: NPV (formula 1, rule 1, adv 0.5, lim 0.5) + IRR (same)

Asked in: Online Preparatory Test (30-09-2026) Q5

Q67. [6 marks] Outlay ₹10,00,000; inflows 3,00,000; 3,50,000; 4,00,000; 2,50,000. Calculate NPV at 12% and estimate IRR by interpolation (use 10% and 12%).

NPV & IRR · ***

Year	Inflow	PVF 12%	PV 12%	PVF 10%	PV 10%
1	3,00,000	0.8929	2,67,870	0.9091	2,72,730
2	3,50,000	0.7972	2,79,020	0.8264	2,89,240
3	4,00,000	0.7118	2,84,720	0.7513	3,00,520
4	2,50,000	0.6355	1,58,875	0.6830	1,70,750
Total			**9,90,485**		**10,33,240**

$$\text{NPV at } 12\% = 9{,}90{,}485 - 10{,}00{,}000 = \mathbf{-9{,}515} \rightarrow \text{reject}$$

$$\text{NPV at } 10\% = +33{,}240$$

$$IRR = 10 + [33{,}240 \div (33{,}240 + 9{,}515)] \times 2 = \mathbf{\approx 11.55\%}$$

IRR 11.55% < 12% → reject (same decision).

Marking scheme: PV table 2 + NPV 1.5 + IRR 2 + decision 0.5

Asked in: Assignment 2 (Capital Budgeting) Q4 and Q5

Q68. [10 marks] Compare NPV and IRR. Why may they conflict for mutually exclusive projects and which one is preferred?

NPV vs IRR · **

Basis	NPV	IRR
Output	Rupee value added	Percentage return
Reinvestment assumption	At cost of capital	At the IRR itself
Rule	NPV > 0	IRR > k
Scale of project	Reflects size	Ignores size
Multiple answers	No	Possible if signs change more than once
Link with wealth	Direct	Indirect

Why conflict?

- Different **size** of investment.

- Different **timing** pattern of inflows.
- Different **life** of projects.
- Different **reinvestment rate** assumptions.

Preference: NPV, because it directly measures the addition to shareholders' wealth and uses a realistic reinvestment rate.

Marking scheme: Table 4 + causes 3 + conclusion 3

Q69. [3 marks] Investment ₹1,00,000; inflows: Y1 30,000, Y2 40,000, Y3 40,000, Y4 30,000, Y5 20,000. Calculate the payback period.

Payback · ***

Year	Inflow	Cumulative
1	30,000	30,000
2	40,000	70,000
3	40,000	1,10,000

```
Payback = E + B/C = 2 + 30,000/40,000
```

Payback = 2.75 years

Marking scheme: Cumulative table 1 + formula 1 + answer 1

Asked in: Online Preparatory Test (30-09-2026) Q3

Q70. [3 marks] Investment ₹6,00,000; equal annual inflow ₹1,50,000 for 6 years; standard payback 4 years. Find the payback period and state whether the project is acceptable.

Payback · ***

```
Payback = 6,00,000 ÷ 1,50,000 = 4 years
```

Equal to the standard period of 4 years, so the project is **just acceptable** (strictly the rule is payback < standard).

Marking scheme: Formula 1 + answer 1 + decision 1

Asked in: Assignment 2 (Capital Budgeting) Q3

Q71. [3 marks] State the merits and demerits of the payback period method.

Payback · **

Merits	Demerits
Simple to understand and calculate	Ignores time value of money (simple form)
Emphasises liquidity; useful when funds are scarce or obsolescence risk is high	Ignores cash flows after the payback period
	Measures speed of recovery, not profitability

Marking scheme: 2 merits + 2 demerits (¾ each)

Q72. [6 marks] Explain the payback period method and the ARR method with merits and demerits.

Payback & ARR · ***

Payback period

```
Payback = Investment ÷ annual inflow | E + B/C (unequal)
```

- **Merits:** simple; stresses liquidity.
- **Demerits:** ignores time value and flows after payback; no profitability measure.

ARR

```
ARR = Average PAT ÷ Average investment × 100
```

- **Merits:** uses accounting data; considers entire life.
- **Demerits:** ignores time value; uses profit, not cash flow.

Marking scheme: Each method: formula 1 + merits 1 + demerits 1

Q73. [3 marks] Distinguish between payback period and ARR (any three points).

Payback vs ARR · ***

Basis	Payback	ARR
Basis	Cash flows	Accounting profit
Measures	Time to recover investment	Rate of return on investment
Rule	Accept if payback < standard	Accept if ARR > cut-off rate
Time value	Ignored (simple form)	Ignored
Strength	Liquidity, simple	Uses whole-life profit

Marking scheme: 3 points × 1

Asked in: Assignment 2 (Capital Budgeting) Q2

Unit V

Q74. [2 marks] What is capital structure?

Capital structure · **

The mix (proportion) of **long-term sources of funds - mainly debt and equity** - used to finance assets and operations. The financing decision seeks the optimum mix that minimises overall cost of capital and maximises firm value.

Marking scheme: Definition 2

Q75. [3 marks] What is the optimum capital structure? Why is it important?

Capital structure · **

The optimum capital structure is the **debt-equity mix that minimises the overall cost of capital and maximises the value of the firm**.

- Too much debt raises financial risk (fixed interest).
- Too much equity raises cost and dilutes control.
- The right mix balances cost, risk and control.

Marking scheme: Meaning 1 + 2 reasons × 1

Q76. [6 marks] Explain the factors affecting capital structure decisions.

Capital structure · **

1. **Cost of capital** of each source.
2. **Risk** - financial risk rises with debt (fixed interest).
3. **Control** - new equity dilutes owners' control.
4. **Flexibility** - ability to raise or change funds later.
5. **Cash-flow ability** to service interest and principal.
6. **Nature of industry, size and stability** of the firm and market conditions.

The aim is the **optimum** structure that minimises overall cost and maximises firm value.

Marking scheme: 6 factors × 1

Q77. [2 marks] What is WACC?

Cost of capital · **

The **Weighted Average Cost of Capital** is the average cost of all sources of long-term finance, each weighted by its proportion in the capital structure.

```
WACC = Σ (Wi × Ki)
```

Marking scheme: Meaning 1 + formula 1

Asked in: Online Preparatory Test (30-09-2026) Q7(c)

Q78. [2 marks] Why is the cost of debt lower than the cost of preference capital?

Cost of capital · ***

- Interest on debt is **tax-deductible**, so the effective cost is I(1 – t).
- Preference dividend is paid out of **post-tax profit**, so there is no tax shield.
- Debt has a prior claim and lower risk for the investor.

Marking scheme: Any 2 points × 1

Q79. [2 marks] State the CAPM formula and explain its terms.

Cost of capital · ***

```
Ke = Rf + β × (Rm − Rf)
```

- Rf = risk-free rate; Rm = market return; β = beta (systematic risk); (Rm – Rf) = market risk premium.

Marking scheme: Formula 1 + terms 1

Asked in: Online Preparatory Test (30-09-2026) Q7(a)

Q80. [2 marks] What is the marginal cost of capital (MCC)?

Cost of capital · **

The **cost of the last (next) rupee of additional capital** raised. It usually rises as more capital is raised; a project should be accepted only if its return exceeds the MCC.

Marking scheme: Meaning 1.5 + use 0.5

Q81. [3 marks] Find the cost of 10% debentures (face ₹100, issued at par) for tax rate 30%, and the cost of 12% preference shares (issued at par).

Cost of capital · ***

```
Kd = I(1 − t) ÷ NP = 10 × (1 − 0.30) ÷ 100 = 7%

Kp = PD ÷ NP = 12 ÷ 100 = 12%
```

Interest is tax-deductible; preference dividend is not.

Marking scheme: Kd 1.5 + Kp 1 + remark 0.5

Asked in: Online Preparatory Test (30-09-2026) Q7(b)

Q82. [3 marks] Debentures of face ₹100 are issued at ₹95 (net proceeds), redeemable at par after 5 years; interest 10%, tax 30%. Find the cost of debt.

Cost of capital · **

```
Kd = [I(1−t) + (RV−NP)/n] ÷ [(RV+NP)/2]
```

I(1–t) = 7; (RV – NP)/n = 5/5 = 1; (RV + NP)/2 = 97.5

Kd = 8 ÷ 97.5 ≈ 8.2%

Marking scheme: Formula 1 + working 1.5 + answer 0.5

Q83. [6 marks] Shree Ltd. has equity ₹5,00,000, 12% preference shares ₹2,00,000 and 10%

debentures ₹3,00,000. Rf 7%, Rm 13%, β 1.5, tax 30%. Find (a) cost of equity by CAPM, (b) after-tax cost of debt, (c) WACC.

Cost of capital · ***

```
(a) Ke = Rf + β(Rm − Rf) = 7 + 1.5 × (13 − 7) = 16%

(b) Kd = 10% × (1 − 0.30) = 7%
```

Source	Amount	Weight	Cost	W × K
Equity	5,00,000	0.5	16%	8.0
Preference	2,00,000	0.2	12%	2.4
Debentures	3,00,000	0.3	7%	2.1
WACC	10,00,000			**12.5%**

Marking scheme: (a) 2 + (b) 1 + (c) 3

Asked in: Online Preparatory Test (30-09-2026) Q7

Q84. [6 marks] Explain the components of cost of capital: cost of debt, preference capital, equity (dividend, growth and CAPM approaches), retained earnings and WACC.

Cost of capital · ***

Source	Formula	Remark
Debt	Kd = I(1−t)/NP	Tax-deductible, cheapest
Preference	Kp = PD/NP	No tax shield
Equity (dividend price)	Ke = D_1/NP	
Equity (growth)	Ke = D_1/NP + g; g = b × r	Gordon
Equity (earnings price)	Ke = E/NP	
Equity (CAPM)	Ke = Rf + β(Rm − Rf)	Systematic risk
Retained earnings	Kr = Ke(1−t_p)(1−b)	Not free

```
WACC = Σ Wi × Ki  (market value weights)
```

Marginal cost of capital = cost of the next rupee raised.

Marking scheme: Table 4 + WACC 1.5 + MCC 0.5

Q85. [10 marks] Explain the concept of cost of capital and its components. Calculate WACC for a firm with equity ₹5,00,000, 12% preference ₹2,00,000 and 10% debentures ₹3,00,000 (Rf 7%, Rm 13%, β 1.5, tax 30%).

Cost of capital · ***

Concept

Cost of capital is the minimum return a firm must earn on investments to satisfy its suppliers of finance; it is the discount rate for projects.

Components

- Debt: Kd = I(1−t)/NP
- Preference: Kp = PD/NP
- Equity: dividend price, growth (Gordon), earnings price, CAPM
- Retained earnings: Kr = Ke(1−t_p)(1−b)

Calculation

```
Ke = 7 + 1.5 × 6 = 16%; Kd = 10 × 0.7 = 7%; Kp = 12%
```

Source	Amount	W	K	W × K

Equity	5,00,000	0.5	16%	8.0
Preference	2,00,000	0.2	12%	2.4
Debentures	3,00,000	0.3	7%	2.1
WACC	10,00,000			**12.5%**

Use

Projects with return above 12.5% add value; use market value weights unless book value is stated. MCC = cost of next rupee.

Marking scheme: Concept 1.5 + components 3 + calculation 4 + use 1.5

Asked in: Online Preparatory Test (30-09-2026) Q7

Q86. [3 marks] D_1 = ₹5, net proceeds per share ₹50, retention ratio 40%, return on retained earnings 15%. Find the cost of equity by the growth method.

Cost of equity · ***

$$g = b \times r = 0.40 \times 15\% = 6\%$$

$$Ke = D_1 \div NP + g = 5 \div 50 + 6\% = 10\% + 6\% = \mathbf{16\%}$$

Marking scheme: g 1 + formula 1 + answer 1

Q87. [3 marks] EPS ₹8 and net proceeds per share ₹64. Find the cost of equity by the earnings price method.

Cost of equity · **

$$Ke = E \div NP = 8 \div 64 = \mathbf{12.5\%}$$

Earnings-price method treats all earnings as the shareholders' return; it is simple but ignores growth.

Marking scheme: Formula 1.5 + answer 1.5

Q88. [6 marks] Explain the methods of computing the cost of equity capital with formulae and examples.

Cost of equity · ***

Method	Formula	Example
Dividend price	Ke = $D_1 \div NP$	D_1 = 5, NP = 50 → 10%
Growth (Gordon)	Ke = $D_1/NP + g$, g = b × r	10% + 6% = 16%
Earnings price	Ke = E ÷ NP	8 ÷ 64 = 12.5%
CAPM	Ke = Rf + β(Rm – Rf)	7 + 1.5 × 6 = 16%
Realised yield	Based on actual dividends and growth received	Uses past data

Retained earnings are not free: Kr = Ke$(1 - t_p)(1 - b)$. Use the method that fits the data given.

Marking scheme: 5 methods × 1 + remark 1

Q89. [10 marks] Explain the cost of equity (all approaches), cost of retained earnings and WACC. Why is WACC used as the discount rate?

Cost of equity & WACC · ***

Meaning

Cost of capital is the minimum return a firm must earn to satisfy its suppliers of finance.

Cost of equity

- Dividend price: D_1/NP

- Growth: $D_1/NP + g$ with $g = b \times r$
- Earnings price: E/NP
- CAPM: $Rf + \beta(Rm - Rf)$
- Realised yield: from actual dividends and growth

Retained earnings

$Kr = Ke(1 - t_p)(1 - b)$ - not free.

WACC

```
WACC = Σ Wi × Ki (market value weights)
```

Example: Equity 5L at 16%, preference 2L at 12%, debentures 3L at 7% after tax → 8 + 2.4 + 2.1 = 12.5%.

Why used as discount rate?

- A project must earn at least the average cost of all funds to add value.
- Used in NPV; project return above WACC → accept.
- MCC is the cost of the next rupee raised.

Marking scheme: Meaning 1 + equity 3 + retained 1 + WACC 3 + use 2

Asked in: Online Preparatory Test (30-09-2026) Q7

Q90. [2 marks] What are the interest coverage ratio and DSCR?

Coverage ratios · **

- **Interest coverage ratio** = EBIT ÷ Interest - how many times interest is covered.
- **DSCR** = (PAT + Interest + Depreciation) ÷ (Interest + Principal) - cash-based ability to service debt.

Marking scheme: Each 1

Q91. [3 marks] Write short notes on interest coverage ratio and DSCR.

Coverage ratios · **

```
Interest coverage = EBIT ÷ Interest

DSCR = (PAT + Interest + Depreciation) ÷ (Interest + Principal instalment)
```

- Higher ratio = bigger cushion to service debt.
- Lenders usually require a minimum DSCR before long-term loans.

Marking scheme: Each ratio 1.5

Q92. [3 marks] PAT ₹2,00,000; interest ₹50,000; depreciation ₹30,000; principal instalment ₹1,20,000. Calculate DSCR.

Coverage ratios · **

```
DSCR = (PAT + Interest + Depreciation) ÷ (Interest + Principal)
```

= (2,00,000 + 50,000 + 30,000) ÷ (50,000 + 1,20,000) = 2,80,000 ÷ 1,70,000

DSCR ≈ 1.65 (cash is 1.65 times the debt obligation).

Marking scheme: Formula 1 + working 1 + answer 1

Q93. [6 marks] Explain interest coverage ratio and DSCR with their importance for lenders and firms.

Coverage ratios · **

```
Interest coverage = EBIT ÷ Interest | DSCR = (PAT + Interest + Depreciation) ÷ (Interest + Principal)
```

Example: EBIT ₹6,00,000, interest ₹1,00,000 → 6 times. PAT 2,00,000, interest 50,000, depreciation 30,000, principal 1,20,000 → DSCR 1.65.

- Higher ratio = bigger cushion to meet fixed obligations even if profits fall.
- ICR is based on profit (EBIT); DSCR is cash-based and includes principal repayment.
- Lenders set a minimum DSCR before sanctioning long-term loans.
- EBIT-EPS analysis alone is not enough - cash-flow ability must also be checked.

Marking scheme: Formulas 2 + example 1.5 + importance 2.5

Q94. [2 marks] What is the indifference point?

EBIT-EPS · **

The **indifference point** is the level of EBIT at which EPS is **the same under two financing plans**.

- Above it → the plan with more debt gives higher EPS.
- Below it → the equity-heavy plan is better.

Marking scheme: Meaning 1.5 + use 0.5

Q95. [3 marks] EBIT ₹3,00,000; interest ₹50,000; tax 30%; 10,000 equity shares. Calculate EPS.

EBIT-EPS · **

```
EPS = (EBIT − I)(1 − t) ÷ N = (3,00,000 − 50,000) × 0.70 ÷ 10,000
```

EPS = ₹17.50

Marking scheme: Formula 1 + working 1 + answer 1

Q96. [3 marks] Find the financial break-even point: interest ₹50,000, preference dividend ₹21,000, tax 30%.

EBIT-EPS · **

```
Financial BEP = I + PD ÷ (1 − t) = 50,000 + 21,000 ÷ 0.70 = 50,000 + 30,000
```

= ₹80,000 (EBIT at which EPS is zero)

Marking scheme: Formula 1 + working 1 + answer 1

Q97. [6 marks] Explain EBIT-EPS analysis and the indifference point. Plan A: all equity, 10,000 shares. Plan B: 5,000 shares plus ₹1,00,000 interest. Find the indifference EBIT.

EBIT-EPS · ***

EBIT-EPS analysis compares the effect of different financing plans on EPS at different EBIT levels.

```
EPS = (EBIT − I)(1 − t) ÷ N

EBIT ÷ 10,000 = (EBIT − 1,00,000) ÷ 5,000 ⇒ EBIT = 2 × (EBIT − 1,00,000)
```

Indifference EBIT = ₹2,00,000 (at 30% tax EPS = ₹14 under both plans).

- Expected EBIT **above** ₹2,00,000 → Plan B (debt) gives higher EPS.
- Expected EBIT **below** ₹2,00,000 → Plan A (equity) is better.

Marking scheme: Concept 2 + equation 2 + answer 1 + rule 1

Q98. [6 marks] A company has 50,000 equity shares of ₹10, 10% preference shares ₹10,00,000 and 10% debentures ₹10,00,000. Tax is 50%. Calculate EPS at EBIT of ₹5,00,000, ₹3,00,000 and ₹7,00,000, and the financial leverage at ₹5,00,000.

EPS & leverage · ***

Item	EBIT 5,00,000	EBIT 3,00,000	EBIT 7,00,000
Less interest (10%)	1,00,000	1,00,000	1,00,000

EBT	4,00,000	2,00,000	6,00,000
Tax 50%	2,00,000	1,00,000	3,00,000
PAT	2,00,000	1,00,000	3,00,000
Less pref. dividend	1,00,000	1,00,000	1,00,000
Earnings for equity	1,00,000	0	2,00,000
EPS (÷ 50,000)	**₹2**	**Nil**	**₹4**

```
FL = EBIT ÷ [EBIT − I − PD/(1−t)] = 5,00,000 ÷ (5,00,000 − 1,00,000 − 2,00,000) = 2.5
```

EBIT ±40% → EPS moves from ₹2 to Nil / ₹4 (financial risk).

Marking scheme: Table 4 + FL 1.5 + comment 0.5

Asked in: Operating & Financial Leverage practice questions (SBPD) Q14

Q99. [10 marks] Explain the financing decision: capital structure factors, EBIT-EPS analysis with indifference point, coverage ratios and the pecking order theory.

Financing decision · ***

Capital structure

Mix of debt and equity; goal = optimum structure. Factors: cost, risk, control, flexibility, cash-flow ability, industry and market conditions.

EBIT-EPS analysis

```
EPS = (EBIT − I)(1−t) ÷ N
```

Indifference EBIT: equate EPS of two plans. Above it → more debt is better; below it → equity is better.

Example: Plan A (10,000 shares) vs Plan B (5,000 shares + ₹1,00,000 interest) → indifference EBIT = ₹2,00,000.

Cash-flow ability

```
Interest coverage = EBIT/I DSCR = (PAT+I+Dep)/(I+Principal)
```

Pecking order

Internal funds → debt → hybrids → new equity.

Marking scheme: Capital structure 2.5 + EBIT-EPS 4 + coverage 1.5 + pecking order 2

Q100. [2 marks] What is leverage? Name its three types.

Leverage · ***

Leverage is the use of fixed-cost funds or fixed-cost assets to magnify returns to the owners.

- **Operating leverage** (fixed operating costs) - effect of sales on EBIT
- **Financial leverage** (fixed financial charges) - effect of EBIT on EPS
- **Combined leverage** = DOL × DFL - effect of sales on EPS

Marking scheme: Meaning 1 + 3 types 1

Q101. [2 marks] What is trading on equity?

Leverage · **

Using borrowed funds to raise returns to equity shareholders. It is **favourable** when the return on investment is **greater than the cost of debt**, because the extra earnings go to the owners.

Marking scheme: Meaning 1.5 + condition 0.5

Q102. [2 marks] Define operating leverage.

Leverage · ***

Operating leverage arises from **fixed operating costs**; it measures the sensitivity of EBIT to a change in sales.

```
DOL = Contribution ÷ EBIT = % change in EBIT ÷ % change in sales
```

Marking scheme: Meaning 1 + formula 1

Q103. [2 marks] Define financial leverage.

Leverage · ***

Financial leverage arises from **fixed financial charges** (interest, preference dividend); it measures the sensitivity of EPS to a change in EBIT.

```
DFL = EBIT ÷ (EBIT − Interest)
```

Marking scheme: Meaning 1 + formula 1

Q104. [2 marks] Define combined leverage.

Leverage · ***

Combined (total) leverage measures the overall effect of fixed operating and financial costs on EPS for a change in sales.

```
DCL = DOL × DFL = Contribution ÷ (EBIT − Interest)
```

Marking scheme: Meaning 1 + formula 1

Q105. [3 marks] Sales ₹10,00,000; variable cost ₹6,00,000; fixed cost ₹2,00,000; interest ₹50,000. Calculate DOL, DFL and DCL.

Leverage · ***

Contribution = 10,00,000 – 6,00,000 = ₹4,00,000; EBIT = 4,00,000 – 2,00,000 = ₹2,00,000

```
DOL = Contribution ÷ EBIT = 4,00,000 ÷ 2,00,000 = 2
DFL = EBIT ÷ (EBIT − I) = 2,00,000 ÷ 1,50,000 = 1.33
DCL = DOL × DFL = 2 × 1.33 = 2.67
```

Marking scheme: Contribution/EBIT 1 + DOL 0.5 + DFL 0.75 + DCL 0.75

Q106. [3 marks] Sales ₹2,50,000; variable cost ₹1,70,000; fixed cost ₹40,000; 12% debt of ₹1,00,000. Find operating, financial and combined leverage.

Leverage · ***

Contribution = 80,000; EBIT = 40,000; interest = 12,000; EBT = 28,000

```
OL = 80,000 ÷ 40,000 = 2
FL = 40,000 ÷ 28,000 = 1.43
CL = 2 × 1.43 = 2.86
```

Marking scheme: Contribution/EBIT 1 + each leverage ⅔

Asked in: Operating & Financial Leverage practice questions (SBPD) Q29

Q107. [3 marks] Sales ₹10,00,000; EBIT is ₹2,00,000 and DOL is 3. If sales rise 10%, by what percentage will EBIT rise?

Leverage · **

```
% change in EBIT = DOL × % change in sales = 3 × 10%
```

EBIT rises by 30% (₹2,00,000 → ₹2,60,000).

Marking scheme: Formula 1.5 + answer 1.5

Q108. [3 marks] Selling price ₹15 per unit, variable cost ₹10, fixed cost ₹10,000. Calculate the operating leverage at 4,000 units and 5,000 units.

Leverage · ***

Units	Contribution (₹5/unit)	EBIT (– 10,000)	DOL = C ÷ EBIT
4,000	20,000	10,000	**2**
5,000	25,000	15,000	**1.67**

As sales rise further above break-even, DOL falls.

Marking scheme: 2 × 1.5

Asked in: Operating & Financial Leverage practice questions (SBPD) Q1

Q109. [3 marks] 3,00,000 units are sold at ₹10; variable cost ₹6; fixed cost ₹4,00,000. If sales rise 10%, by what percentage will EBIT rise?

Leverage · ***

Contribution = 3,00,000 × 4 = ₹12,00,000; EBIT = 12,00,000 – 4,00,000 = ₹8,00,000

```
DOL = 12,00,000 ÷ 8,00,000 = 1.5
% rise in EBIT = 1.5 × 10% = 15%
```

Marking scheme: Contribution/EBIT 1 + DOL 1 + answer 1

Asked in: Operating & Financial Leverage practice questions (SBPD) Q5

Q110. [6 marks] Explain operating, financial and combined leverage with formulae and an example.

Leverage · ***

Leverage	Arises from	Measures	Formula
Operating	Fixed operating costs	Effect of sales on EBIT	DOL = Contribution ÷ EBIT
Financial	Fixed financial charges (interest, pref. dividend)	Effect of EBIT on EPS	DFL = EBIT ÷ (EBIT – I)
Combined	Both	Effect of sales on EPS	DCL = DOL × DFL

Example: Sales 10,00,000; VC 6,00,000; FC 2,00,000; interest 50,000 → Contribution 4,00,000; EBIT 2,00,000; DOL = 2; DFL = 1.33; DCL = 2.67.

- High DOL → high business risk; high DFL → high financial risk.
- Favourable financial leverage (trading on equity) when return on investment > cost of debt.

Marking scheme: Table 3 + example 2 + comments 1

Q111. [6 marks] Sales ₹5,00,000; variable cost 40% of sales; fixed cost ₹2,00,000; interest ₹60,000. Calculate OL, FL and CL and the % rise in profit before tax if sales rise 10%.

Leverage · ***

VC = 2,00,000; Contribution = 3,00,000; EBIT = 1,00,000; EBT = 40,000

```
OL = 3,00,000 ÷ 1,00,000 = 3
FL = 1,00,000 ÷ 40,000 = 2.5
```

```
CL = 3 × 2.5 = 7.5
```

% rise in PBT = CL × % rise in sales = 7.5 × 10% = **75%**.

Marking scheme: Basics 1 + OL/FL/CL 3 + % rise 2

Asked in: Operating & Financial Leverage practice questions (SBPD) Q32

Q112. [6 marks] A firm expects EBIT of ₹30,000, has 5% debentures of ₹1,20,000, 10% preference shares of ₹60,000 and 3,000 equity shares of ₹10. Tax is 50%. Find EPS and the degree of financial leverage at EBIT ₹30,000, and the EPS at EBIT ₹18,000 and ₹42,000.

Leverage · ***

Interest = 5% × 1,20,000 = ₹6,000; preference dividend = ₹6,000 (pre-tax equivalent = 6,000 ÷ 0.5 = 12,000)

EBIT	EBT	PAT	Less PD	EPS (÷ 3,000)
30,000 (base)	24,000	12,000	6,000	₹2
18,000	12,000	6,000	6,000	Nil
42,000	36,000	18,000	6,000	₹4

```
DFL (EBIT 30,000) = 30,000 ÷ (30,000 − 6,000 − 12,000) = 2.5
```

EBIT falls 40% → EPS falls 100% (unfavourable); EBIT rises 40% → EPS doubles (favourable).

Marking scheme: Table 3.5 + DFL 1.5 + comment 1

Asked in: Operating & Financial Leverage practice questions (SBPD) Q16

Q113. [10 marks] Explain operating, financial and combined leverage. Calculate DOL, DFL and DCL for: Sales ₹10,00,000, variable cost ₹6,00,000, fixed cost ₹2,00,000, interest ₹50,000. What do the results mean?

Leverage · ***

Concepts

Type	Source	Effect	Formula
Operating	Fixed operating cost	Sales → EBIT	DOL = C ÷ EBIT
Financial	Interest / pref. dividend	EBIT → EPS	DFL = EBIT ÷ (EBIT–I)
Combined	Both	Sales → EPS	DCL = DOL × DFL

Calculation

Item	Working	₹
Contribution	10,00,000 – 6,00,000	4,00,000
EBIT	4,00,000 – 2,00,000	2,00,000
EBT	2,00,000 – 50,000	1,50,000
DOL	4,00,000 ÷ 2,00,000	2
DFL	2,00,000 ÷ 1,50,000	1.33
DCL	2 × 1.33	2.67

Interpretation

- Sales +10% → EBIT +20% (DOL 2).
- EBIT +10% → EPS +13.3% (DFL 1.33).
- Sales +10% → EPS +26.7% (DCL 2.67).
- Higher leverage = higher return in good times and higher risk in bad times.

Marking scheme: Concepts 3 + calculation 4 + interpretation 3

Q114. [2 marks] What is the pecking order theory of financing?

Pecking order · **

Firms follow a **hierarchy of finance** (Myers and Majluf): **internal funds → debt → hybrid securities → new equity** (last resort).

Marking scheme: Statement 1 + order 1

Q115. [3 marks] Why do firms prefer internal funds and debt to new equity? (pecking order)

Pecking order · **

1. **Internal funds** have no issue cost, no information asymmetry and no dilution of control.
2. **Debt** is the next safe source - no dilution and signals confidence.
3. **New equity** is the last resort: the market may read it as the shares being overvalued, so the price falls.

Marking scheme: 3 × 1

Q116. [6 marks] Explain the pecking order theory of financing.

Pecking order · **

Proposed by Myers and Majluf: firms have no fixed target debt ratio; they follow a hierarchy because of information asymmetry and signalling costs.

1. **Internal funds** (retained earnings) - no flotation cost, no dilution.
2. **Debt** - no dilution; signals confidence.
3. **Hybrid securities** (convertible debentures).
4. **New equity** - last resort; market may read it as overvaluation.

It explains why profitable firms use less debt - they have enough internal funds.

Marking scheme: Statement 1.5 + 4 steps 3 + note 1.5

Q117. [3 marks] A firm uses 60% equity costing 15% and 40% debt costing 8% after tax. Find the WACC.

WACC · ***

Source	Weight	Cost	W × K
Equity	0.6	15%	9.0
Debt	0.4	8%	3.2
WACC			**12.2%**

Marking scheme: Weights 1 + products 1 + WACC 1